

```
1 `timescale 1ns / 1ps
2 module TimerCounter(
3     input CLK,
4     input [31:0] top_in,
5     input [31:0] cmp_in,
6     output PWM
7 );
8     reg [31:0] cnt;
9     reg [31:0] top;
10    reg [31:0] cmp;
11    always @(posedge CLK)
12    begin : CNT_MOD
13        cnt <= cnt + 1;
14        if(cnt >= top)
15            cnt <= 0;
16    end
17
18    assign PWM = (cnt < cmp);
19
20    always @(posedge CLK)
21    begin : TOP_MOD
22        top <= top_in;
23    end
24
25    always @(posedge CLK)
26    begin : CMP_MOD
27        cmp <= cmp_in;
28    end
29 endmodule
```